Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Develop Multi-Agent AI Applications with AutoGen

TGS Ref No: TGS-2024045806

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 9

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
8.0	1 March 2024	Legacy release — LLM orchestration with LangChain/LangGraph, RAG, and multi-agent frameworks (AutoGen, CrewAI).	Dr. Alfred Ang
9	28 August 2026	Full revamp on Microsoft AutoGen AgentChat 0.7: uv-based Pre-Lab, 8 AutoGen labs (agents, model clients, tools, ChromaDB RAG with evaluation, ReAct, human-in-the-loop, RoundRobin and Selector teams), slide references re-mapped to the v9 deck.	Dr. Alfred Ang

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

TABLE OF CONTENTS

Course Information **4**
Learning Outcomes **4**
Assessment **4**
Course Schedule **4**
Day 1 — LLM Orchestration & RAG with AutoGen 4
Day 2 — Multi-Agent Workflows & Assessment 5
Lab Reference (aligned to learning units) **6**

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Course Information

Course Title	Develop Multi-Agent AI Applications with AutoGen
WSQ Course Reference	TGS-2024045806
TSC	Artificial Intelligence Application (AER-TEM-4026-1.1)
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 8 hours per day (16 hours, including 2 hours of assessment)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on AutoGen labs in a uv-managed Python environment
Trainer	Dr. Alfred Ang
Slide Deck	Develop Multi-Agent AI Applications with AutoGen-v9.pptx (124 slides — slide numbers below refer to this deck)

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Evaluate Large Language Model (LLM) AI models by identifying their strengths and limitations.
- LO2: Analyze Retrieval-Augmented Generation (RAG) algorithms to improve efficiency.
- LO3: Assess the feasibility of implementing multi-agent AI applications.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on AutoGen build tasks, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 2 from 4:30 pm (briefing 4:15 pm).
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Instructional methods: Lecture, Didactic Questioning, Demonstration and Practical (hands-on labs). Slide references are to the v9 trainer deck.

Day 1 — LLM Orchestration & RAG with AutoGen

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, digital attendance (AM), trainer and learner introductions, ground rules, course material download, learning outcomes and course

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

		outline (Slides 2–18). Pre-Lab check: uv environment ready (Lab 0)
10:00–11:00	60 min	**Core concepts — from LLMs to multi-agent AI: how LLMs work, strengths and limitations, agents and the agentic loop, the AutoGen stack (Slides 19–38)**
11:00–11:15	15 min	Tea break
11:15–12:15	60 min	**Topic 1 — Introduction to LLM AI Orchestration: key concepts, local LLMs, building and debugging an LLM app, function calling (Slides 39–48)**
12:15–13:15	60 min	Hands-on: Lab 1: Set Up AutoGen and Run Your First Agent (Slides 49–53) and Lab 2: Configure Model Clients and Structured Output begins (Slides 54–58)
13:15–14:15	60 min	Lunch break
14:15–15:45	90 min	Digital attendance (PM). Hands-on: Lab 2 completes; Lab 3: Equip an Agent with Tools (Slides 59–64)
15:45–16:00	15 min	Tea break
16:00–17:00	60 min	**Topic 2 — Retrieval-Augmented Generation: pipeline, embeddings, chunking, vector databases, similarity search, agentic RAG, evaluation (Slides 65–76)**
17:00–18:15	75 min	Hands-on: Lab 4: Build a Knowledge Index with Embeddings and a Vector Database (Slides 77–81) and Lab 5: Build and Evaluate a Grounded RAG Agent (Slides 82–87)
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 2 — Multi-Agent Workflows & Assessment

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)
9:45–10:45	60 min	**Topic 3 — Implementing a Multi-Agent AI Workflow: ReAct, conversation patterns, termination conditions, human-in-the-loop (Slides 88–99)**
10:45–11:00	15 min	Tea break

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

11:00–12:15	75 min	Hands-on: Lab 6: Implement a ReAct Agent with Human-in-the-Loop (Slides 100–104)
12:15–13:15	60 min	Hands-on: Lab 7: Orchestrate a Round-Robin Multi-Agent Team (Slides 105–109)
13:15–14:15	60 min	Lunch break
14:15–15:30	75 min	Digital attendance (PM). Hands-on: Lab 8: Route Specialists with SelectorGroupChat (Capstone) (Slides 110–115), incl. feasibility readiness checks and AutoGen Studio
15:30–15:45	15 min	Tea break
15:45–16:15	30 min	Revision and course summary (Slides 116–120), course feedback and TRAQOM survey
16:15–16:30	15 min	**Briefing for Assessment (Slide 13) and digital attendance (Assessment)**
16:30–17:30	60 min	**Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book**
17:30–18:30	60 min	**Practical Performance (PP) — hands-on AutoGen build tasks, 1 hour, open book**

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to learning units)

Topic / Learning unit	Weighting	Labs (slide references)
Topic 01: Introduction to LLM AI Orchestration	36%	Lab 1 (Slides 49–53); Lab 2 (Slides 54–58); Lab 3 (Slides 59–64)
Topic 02: Retrieval-Augmented Generation (RAG)	36%	Lab 4 (Slides 77–81); Lab 5 (Slides 82–87)
Topic 03: Implementing a Multi-Agent AI Workflow	28%	Lab 6 (Slides 100–104); Lab 7 (Slides 105–109); Lab 8 (Slides 110–115)

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg